

Time-Error-Constrained Bidirectional Provisioning for Synchronization Services in Dynamic Elastic Optical Networks

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Abstract *Bidirectional slot mismatch in dynamic EONs induces chromatic-dispersion time errors that can invalidate synchronization services. We derive a path-dependent reverse-slot admissibility bound for bidirectional provisioning. On COST239, it lowers blocking versus timing-unaware and strict same-slot baselines while keeping admitted synchronization services within a 3-ns budget. ©2026 The Author(s)*

Introduction

Emerging deployments such as disaggregated radio access networks, industrial 5G and time-sensitive networking (TSN) systems, and wide-area optical timing networks increasingly require deterministic and high-precision timing support over shared and reconfigurable optical transport networks [1–3]. To realize such optical synchronization, packet-based protocols such as the Precision Time Protocol (PTP) and White Rabbit (WR) rely on bidirectional reciprocity for accurate clock-offset estimation and common-mode delay cancellation [4, 5]. Historically, this reciprocity requirement was typically satisfied over dedicated fiber or statically engineered wavelength pairs, where physical asymmetry could be measured and compensated under relatively stable operating conditions [3, 5]. As elastic optical networks (EONs) have emerged as a key shared and reconfigurable optical-transport substrate, reciprocity preservation increasingly becomes a dynamic provisioning issue rather than only a static calibration issue.

In such dynamic EONs, the forward and reverse directions of a bidirectional lightpath may be assigned different spectrum slots under contention and fragmentation. Because chromatic dispersion makes group delay frequency-dependent, such forward–reverse slot mismatch introduces delay asymmetry that biases reciprocal timing estimation [6, 7]. Synchronization feasibility is therefore not a property of a route alone, but of the coupled forward–reverse spectrum pair selected on that route. Since reprovisioning can change this pairing, a lightpath may remain spectrally and QoT-feasible yet still violate the allowable time error (TE) budget.

To the best of our knowledge, prior work has not explicitly translated synchronization reciprocity into an operational constraint for dynamic bidirectional spectrum assignment in EONs: optical time/frequency transfer studies mainly address calibration, compensation, and link modeling under dedicated or quasi-static conditions [3, 5–7], whereas conventional dynamic routing, modulation, and

spectrum assignment (RMSA) primarily optimizes QoT and spectrum efficiency in a timing-unaware manner [8–10]. This leaves two representative operational extremes: Strict Same-Slot provisioning, which removes CD-induced mismatch at the cost of spectral rigidity, and timing-unaware dynamic RMSA, which preserves flexibility but may admit synchronization-infeasible lightpaths under nanosecond-level TE budgets. The resulting challenge is to preserve synchronization feasibility without reverting to rigid slot symmetry.

To address this challenge, we propose a time-error-aware (TE-aware) bidirectional RMSA framework that translates a continuous physical-layer TE budget into a discrete, path-dependent reverse-slot admissibility bound. By pruning only reverse-slot candidates that are physically incompatible with the timing requirement, the proposed method suppresses configuration-dependent CD-induced asymmetry while retaining spectrum flexibility. Numerical results on the COST239 topology show that, at 800 Erlangs, the proposed approach reduces overall blocking by 74.6% and 28.0% relative to timing-unaware and strict same-slot baselines, respectively, while keeping admitted synchronization services within the 3 ns TE budget.

Time-Error-Constrained Bidirectional Provisioning

In the proposed framework, the physical-layer TE budget is translated into a path-dependent reverse-slot search window for bidirectional provisioning, as illustrated in Fig. 1. Instead of enforcing rigid same-slot symmetry, this rule allows bounded forward–reverse slot offsets whenever they satisfy the synchronization requirement.

Let the forward and reverse center frequencies be f_c^{fwd} and f_c^{rev} , respectively, and define the forward–reverse frequency offset as $\Delta f = |f_c^{\text{fwd}} - f_c^{\text{rev}}|$. Using a first-order approximation around the reference wavelength λ_0 , the corresponding wavelength offset is $\Delta\lambda \approx \frac{\lambda_0^2}{c} \Delta f$, where c is the speed of light. Under the route-reciprocity assumption, both directions share the same effec-

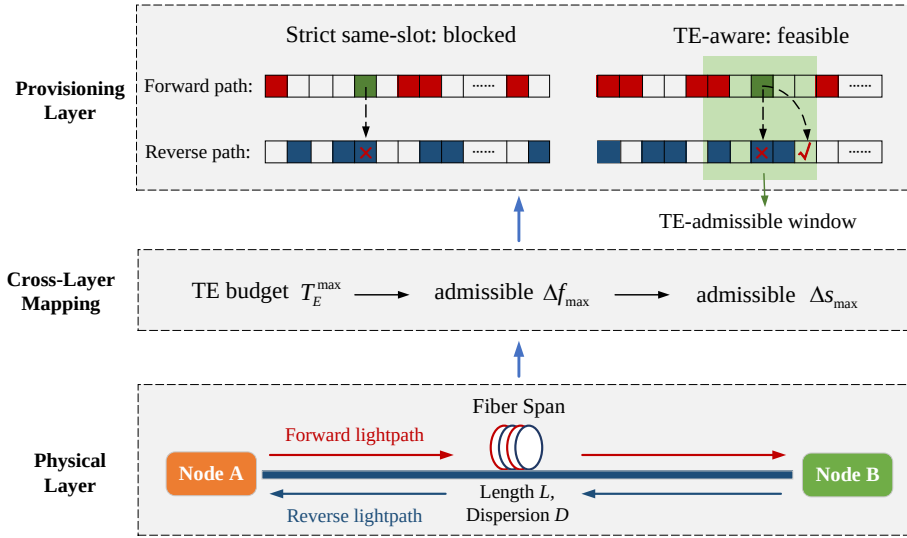


Fig. 1: Cross-layer translation from the time-error budget to a TE-admissible reverse-slot window for bidirectional provisioning.

tive path length L for TE estimation, and the total synchronization error is modeled as

$$TE_{\text{total}} = \frac{1}{2}DL|\Delta\lambda| + h\delta_{\text{node}}, \quad (1)$$

where D is the chromatic-dispersion coefficient, h is the hop count, and δ_{node} is a lumped residual per-hop asymmetry term capturing slowly varying or non-configurable frequency-independent biases. The factor $1/2$ follows from two-way time-transfer estimation, in which the synchronization error equals half of the forward–reverse one-way delay difference.

For a synchronization request with TE budget T_E^{max} , feasible bidirectional provisioning requires $TE_{\text{total}} \leq T_E^{\text{max}}$. Let Δf_{slot} denote the slot width. When $T_E^{\text{max}} > h\delta_{\text{node}}$, the maximum admissible reverse-slot offset is

$$\Delta s_{\text{max}} = \left\lfloor \frac{c}{\lambda_0^2 \Delta f_{\text{slot}}} \cdot \frac{2(T_E^{\text{max}} - h\delta_{\text{node}})}{DL} \right\rfloor, \quad (2)$$

whereas for $T_E^{\text{max}} \leq h\delta_{\text{node}}$, the admissible radius collapses to zero, so only a zero-offset candidate can be examined and the request is rejected if Eq. (1) is still violated. Equation (2) therefore translates the continuous TE budget into a discrete, path-dependent reverse-slot search radius for online provisioning.

Based on this mapping, the forward direction is first provisioned by a conventional K-shortest-path routing and First-Fit spectrum assignment (KSP-FF) engine. For synchronization services, once the forward slot is fixed, the reverse direction is searched only within the TE-admissible window

$$s^{\text{rev}} \in [s^{\text{fwd}} - \Delta s_{\text{max}}, s^{\text{fwd}} + \Delta s_{\text{max}}], \quad (3)$$

whereas non-synchronization services are provisioned by the same KSP-FF engine without TE filtering. Candidates within the local window are

examined in increasing offset order, and a synchronization request is admitted only if the forward–reverse pair satisfies both $TE_{\text{total}} \leq T_E^{\text{max}}$ and the QoT requirement in both directions. Since L and h are already known once a candidate path is selected, evaluating Δs_{max} is a constant-time operation; the TE-aware extension therefore prunes reverse-slot candidates without changing the asymptotic order of the KSP-FF scan.

Results and Discussion

We evaluate the proposed framework on the COST239 topology. For each node pair, $K = 5$ candidate paths are precomputed, each directed link provides 320 frequency slots (12.5 GHz/slot), and dynamic requests arrive with requested bit rates drawn from $\{50, 100, 200, 400\}$ Gb/s. The synchronization-service ratio is $\rho_{\text{sync}} = 0.3$, the TE budget is fixed at $T_E^{\text{max}} = 3$ ns, and the TE model uses $D = 17$ ps/(nm·km) at $\lambda_0 = 1550$ nm with a lumped residual per-hop margin of $\delta_{\text{node}} = 20$ ps. Each data point is averaged over 10 random seeds (100,000 requests/seed). We compare the proposed TE-Aware scheme against two representative operational extremes: Unaware, where reverse-slot candidates are generated without TE-guided pruning and the TE constraint is checked only at final admission, and Strict Same-Slot, which enforces identical forward–reverse slots.

Fig. 2(a) shows the physical admissibility structure underlying the proposed method. The maximum admissible reverse-slot radius Δs_{max} decreases as path length increases and as the TE budget tightens, indicating that a fixed empirical reverse-slot rule is generally inadequate: long paths approach strict same-slot behavior, whereas short paths still retain limited reverse-slot flexibility.

Fig. 2(b) reports the 95th-percentile synchronization time error of admitted synchronization services. Strict Same-Slot provides a conservative

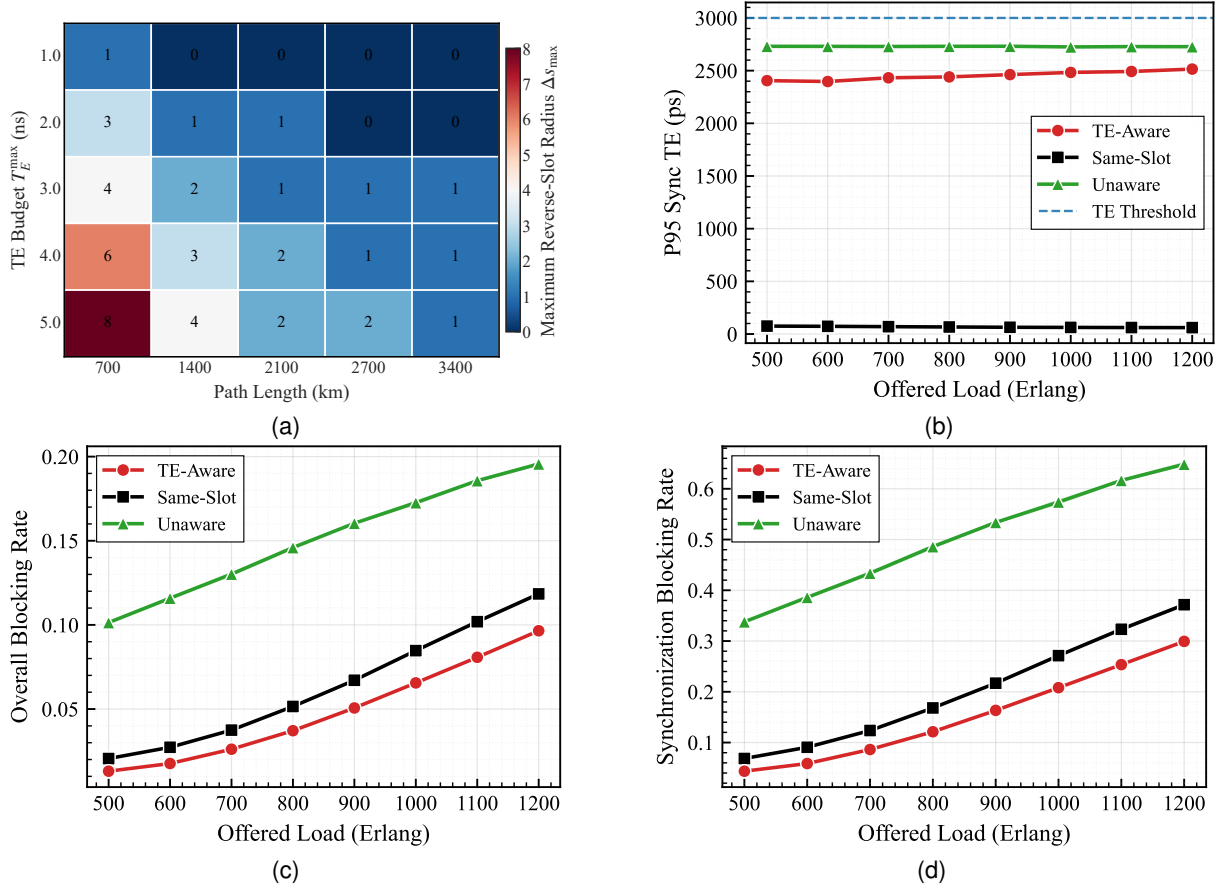


Fig. 2: Performance in COST239 under dynamic traffic: (a) maximum admissible reverse-slot radius $\Delta s_{\max}$ versus path length and TE budget; (b) 95th-percentile synchronization time error of admitted synchronization services; (c) overall blocking rate; and (d) synchronization-service blocking rate. Panels (b)–(d) are plotted versus offered load. Shaded bands denote ± 1 standard deviation and are negligible at the plotting scale in most cases.

lower-TE reference, whereas Unaware operates much closer to the 3-ns limit (around 2730 ps), leaving limited timing margin among admitted services. TE-Aware reaches about 2515 ps at 1200 Erlangs while remaining within the TE budget, showing that synchronization feasibility can be preserved without collapsing to rigid slot symmetry.

Figs. 2(c) and (d) show the corresponding blocking performance. Across the examined load range, TE-Aware achieves the lowest overall and synchronization-service blocking among the three schemes. At 800 Erlangs, the overall blocking rates are 0.1459 for Unaware, 0.0515 for Strict Same-Slot, and 0.0371 for TE-Aware, corresponding to reductions of 74.6% and 28.0% relative to Unaware and Strict Same-Slot, respectively. For synchronization services, the corresponding blocking rates are 0.4860, 0.1682, and 0.1212, again confirming the lowest blocking for TE-Aware.

The reason is that nominal spectrum freedom does not necessarily translate into effective admission capacity. In Unaware, highly mismatched forward–reverse pairings may still be explored but are later rejected by the TE check, whereas Strict Same-Slot eliminates CD-induced mismatch at the cost of useful flexibility under fragmentation. TE-

Aware instead prunes only reverse-slot candidates that are physically incompatible with the timing requirement, thereby preserving effective spectrum flexibility while avoiding synchronization-infeasible pairings.

Conclusion

We proposed a time-error-aware (TE-aware) bidirectional RMSA framework for synchronization-aware provisioning in dynamic EONs. Under a CD-dominated and route-reciprocal TE model, the proposed method translates a continuous TE budget into a discrete, path-dependent reverse-slot admissibility bound, thereby operationalizing synchronization feasibility during online bidirectional provisioning. Numerical results on COST239 show that, at 800 Erlangs, the proposed scheme reduces overall blocking by 74.6% and 28.0% relative to timing-unaware and strict same-slot baselines, respectively, while keeping admitted synchronization services within the 3 ns TE budget. These results indicate that cross-layer TE-aware provisioning can suppress configuration-dependent CD mismatch while preserving useful spectrum flexibility, without reverting to rigid slot symmetry. Future work will incorporate richer hardware-level asymmetries and broader topology evaluations.

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